Disappearance of Spurious Valleys via Overparameterization

Márk Penc

ETH Zurich, Switzerland

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Advisor: Valentin Abadie
Host group: Chair for Mathematical Information Science
Prof. Dr. Helmut Bölcskei

Outline

1. Problem Formulation

2. Prior Work

3. Main Result

4. Additional Results
4.1 Tightness
4.2 Probabilistic Escape

5. Recap & Discussion

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Problem Formulation

Goal:

- Understand geometry of the loss

Why?

- **Optimization** methods $\rightarrow$ find good hypothesis **via loss**
- **Geometry** of loss surfaces:
  - **non-convex**
  - many non-global, **local minima**
  - admit **complex structures**

$\Rightarrow$ Fundamental mathematical understanding is important

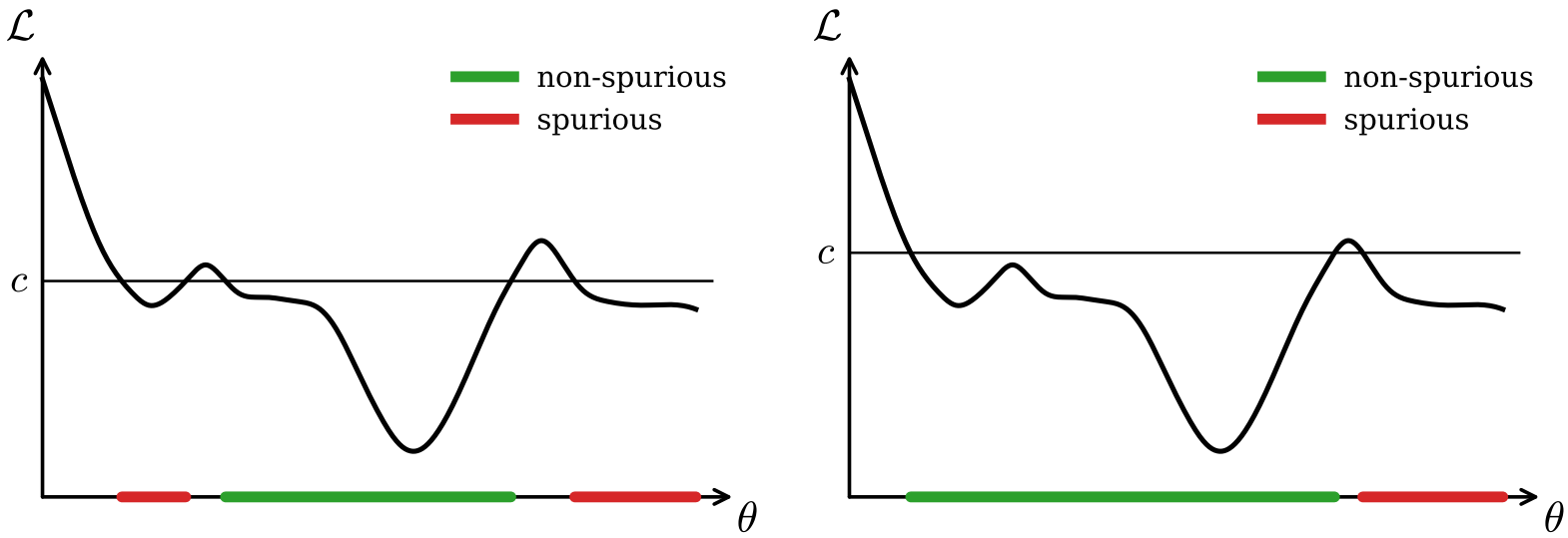
Main Focus:

Spurious Valleys

Problem Formulation

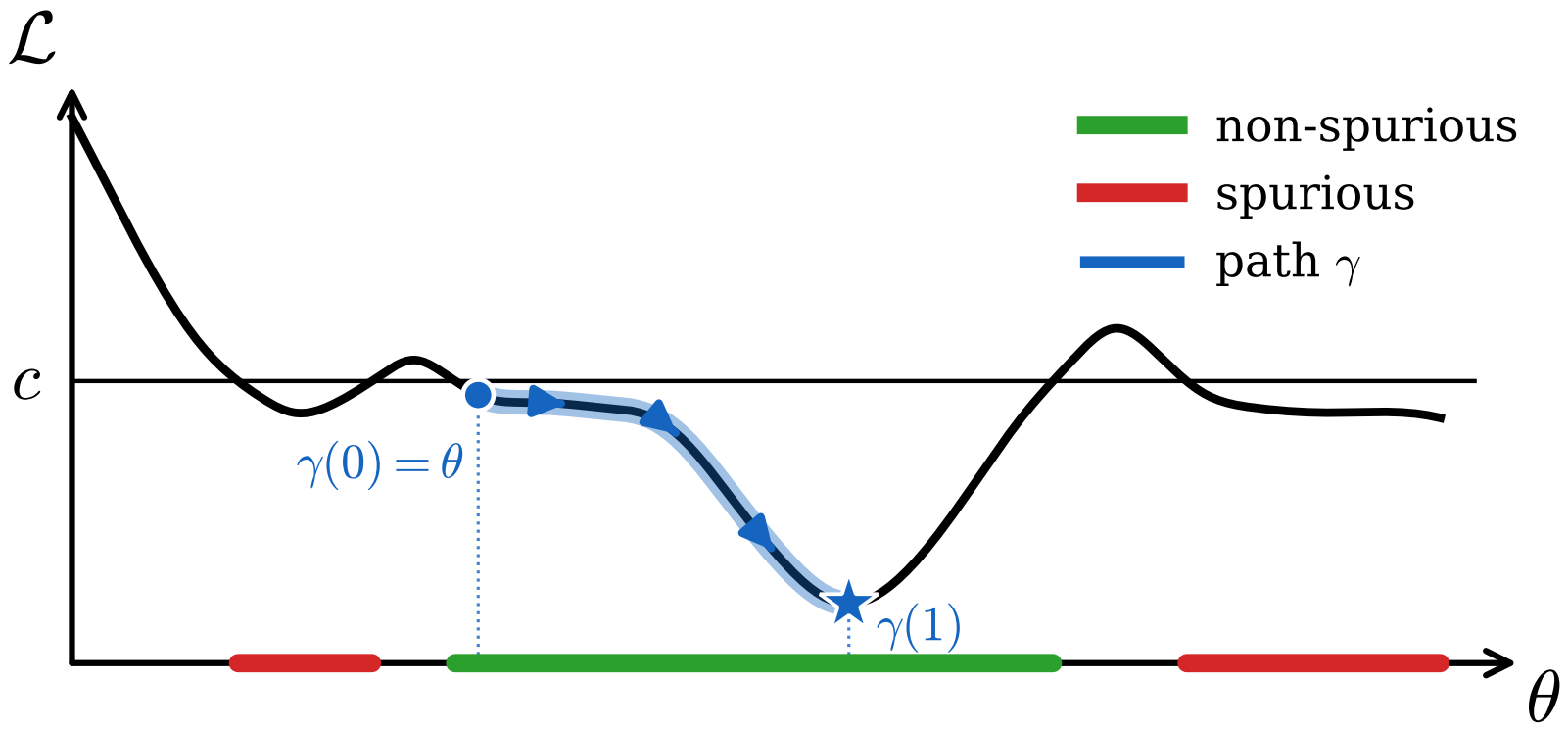
Spurious Valleys:

- Connected sub-level sets that do not contain a global minimum



Problem Formulation

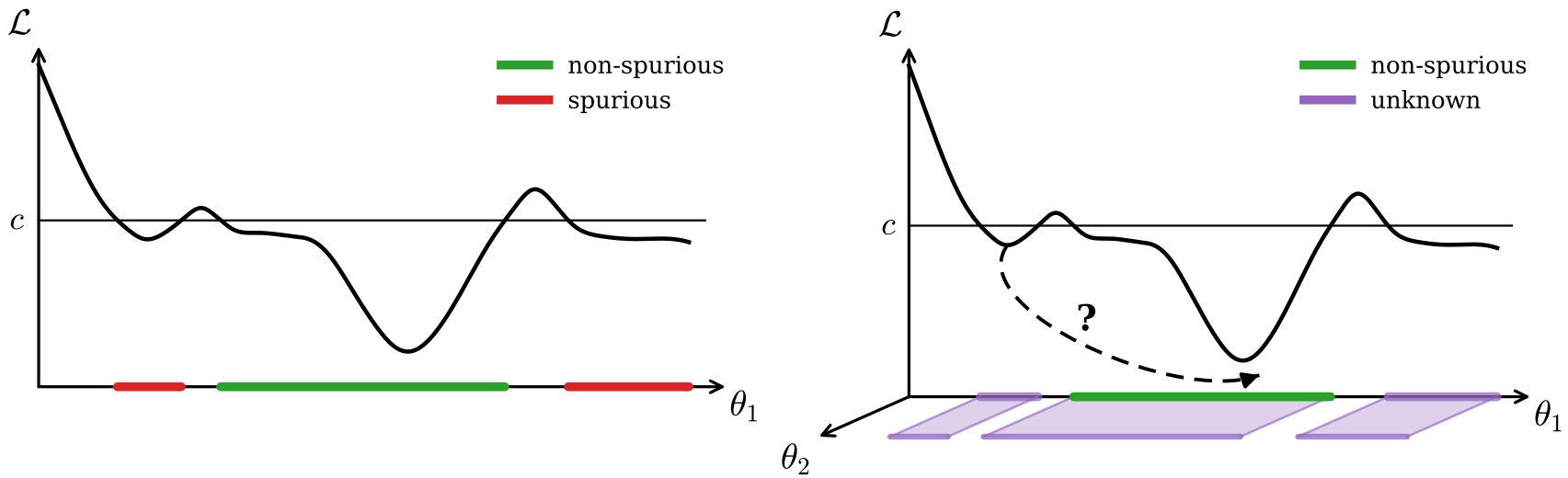
- **Non-spurious**: non-increasing path to global minimum
- **Spurious**: every escape must first go uphill



Problem Formulation

Main Question:

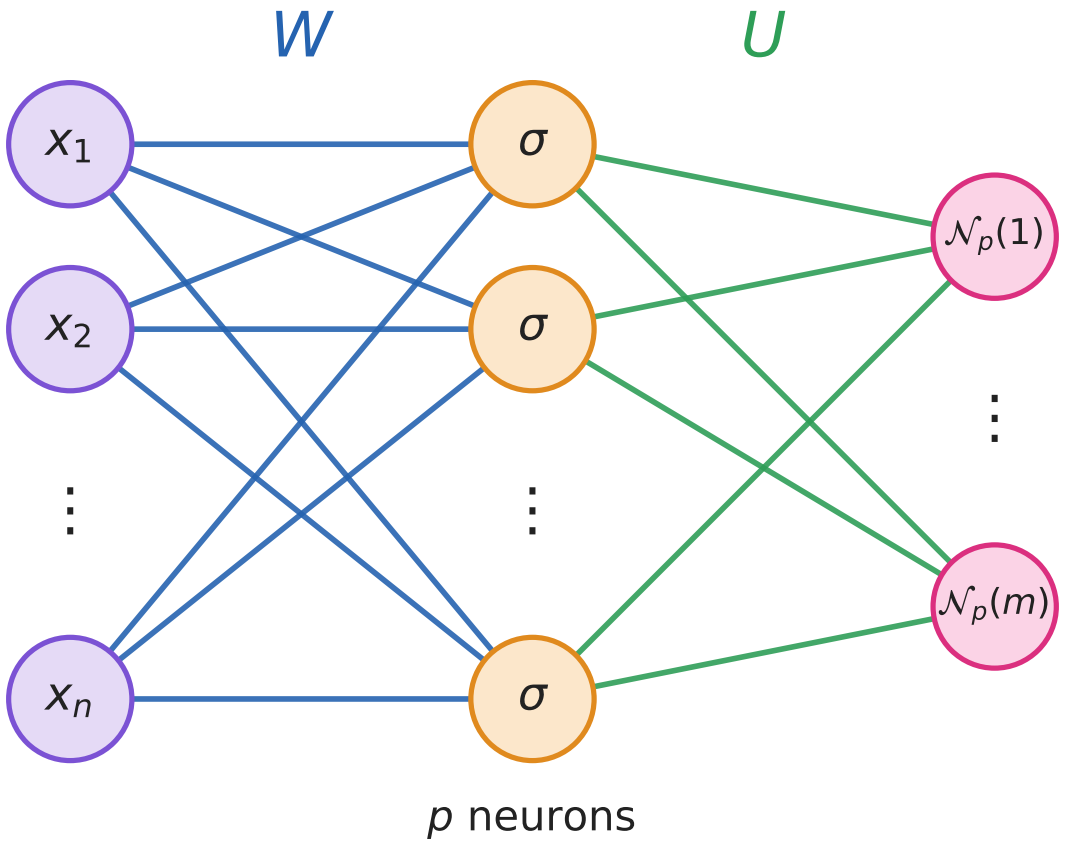
Do spurious valleys disappear via overparameterization?



Problem Formulation - Neural Networks

Width p neural network $\mathcal{N}_p : [0,1]^n \times \Theta_p \to \mathbb{R}^m$:

$$\mathcal{N}_p(x;\theta) = U\,\vec{\sigma}(Wx)$$



Problem Formulation - Loss Surface

- **Functional loss** on a function class $\mathcal{F}$:

$$l : \mathcal{F} \to \mathbb{R}_{\geq 0},$$

  convex, and **continuous**

- Induced **parameter loss** $\mathcal{L}_p : \Theta_p \to \mathbb{R}_{\geq 0}$:

$$\mathcal{L}_p(\theta) = l(\mathcal{N}_p(\cdot; \theta))$$

Outline

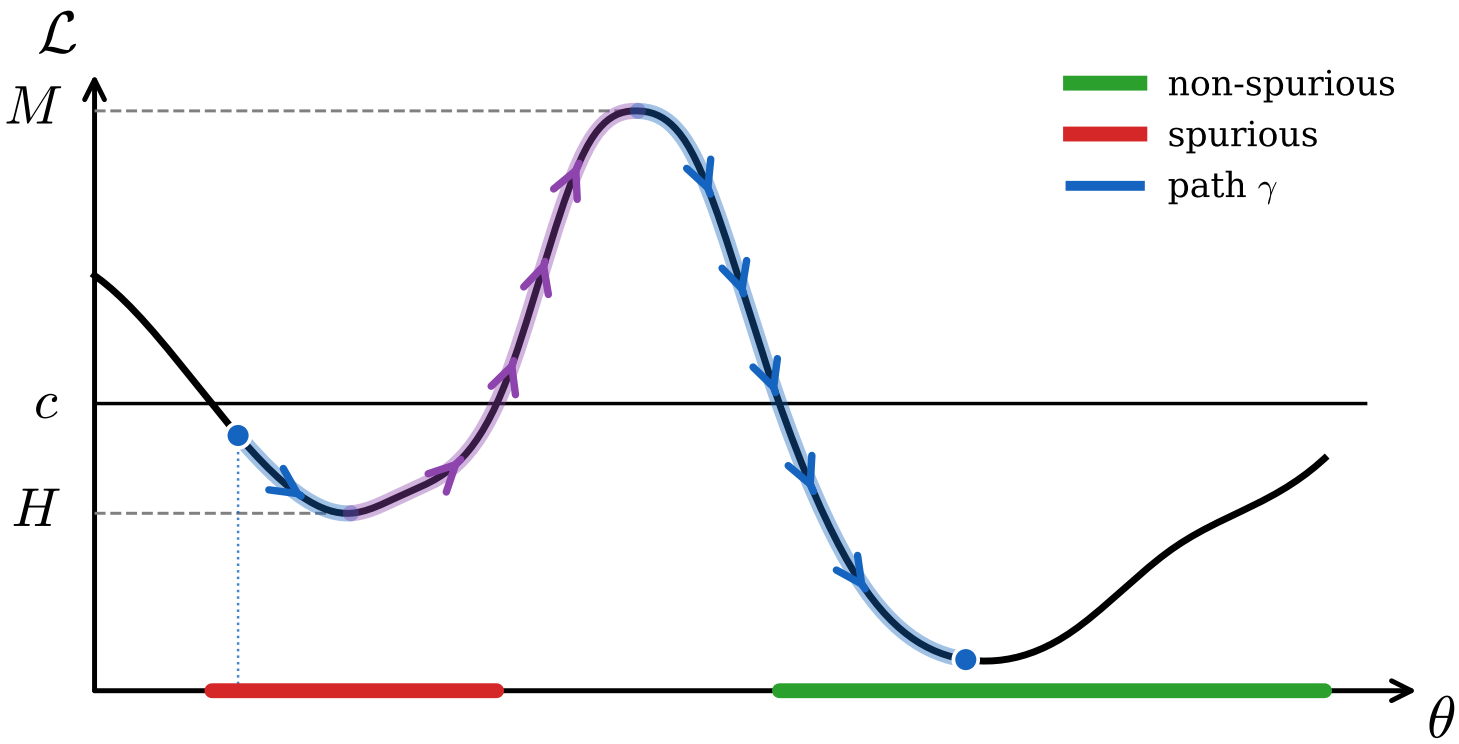
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Prior Work - Spurious Valleys

[Venturi, Bandeira & Bruna (2019)] Corollary 15, informally

For common architectures:

- **arbitrarily bad spurious valleys may exist**
- far away from glob. min. + have to climb up

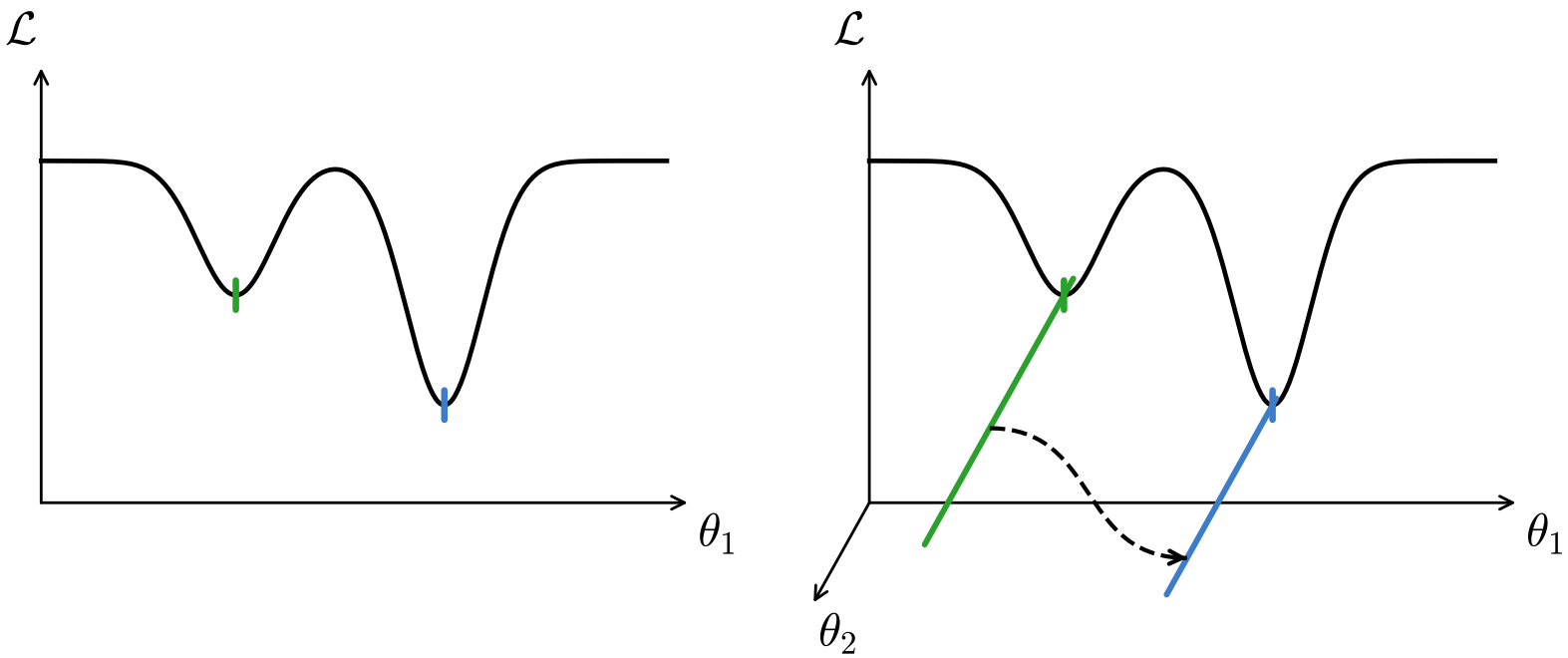


Extra Dimensions $\rightarrow$ Degrees of Freedom

[Şimşek et al. (2021), Farrugia-Roberts (2022)]

How do extra dimensions help?

$\Longrightarrow$ additional parameters where the same function is implemented



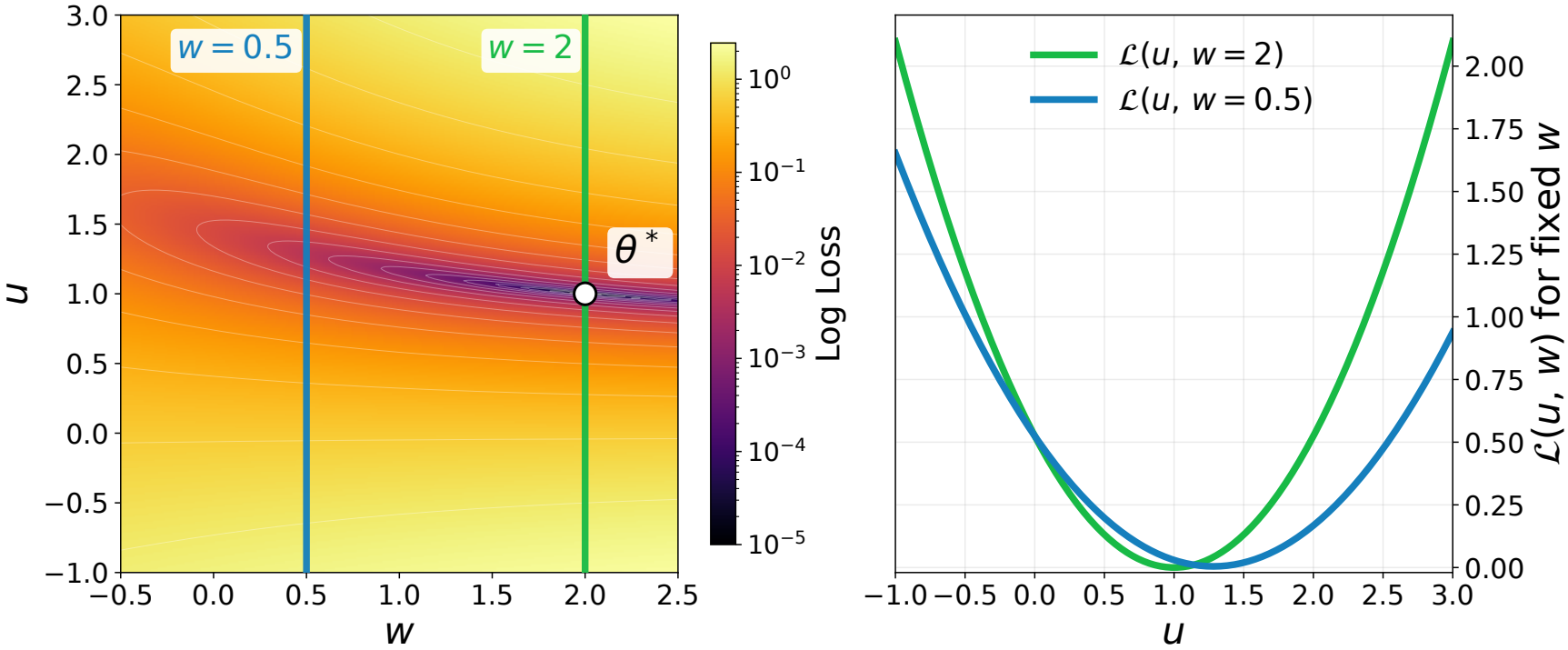
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Convexity in Parameter Space

Proposition

When two parameters share W, the loss of their linear interpolation is **convex in parameter space**



Question

So can we escape?

Question

So can we escape?

Answer

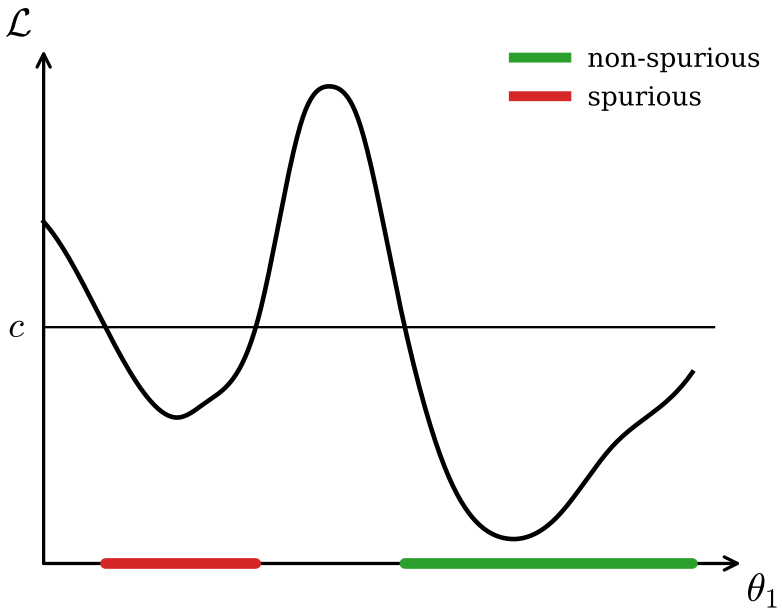
Yes!

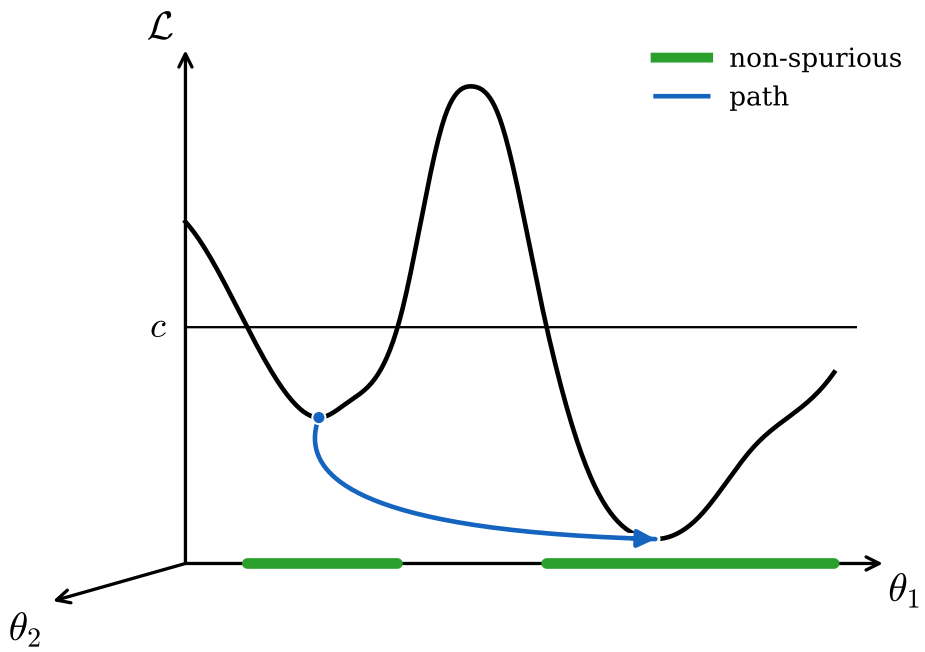
Main Result

Theorem (informally)

Under mild assumptions:

- **All** spurious valleys disappear via sufficient overparameterization
- Doubling the width is enough

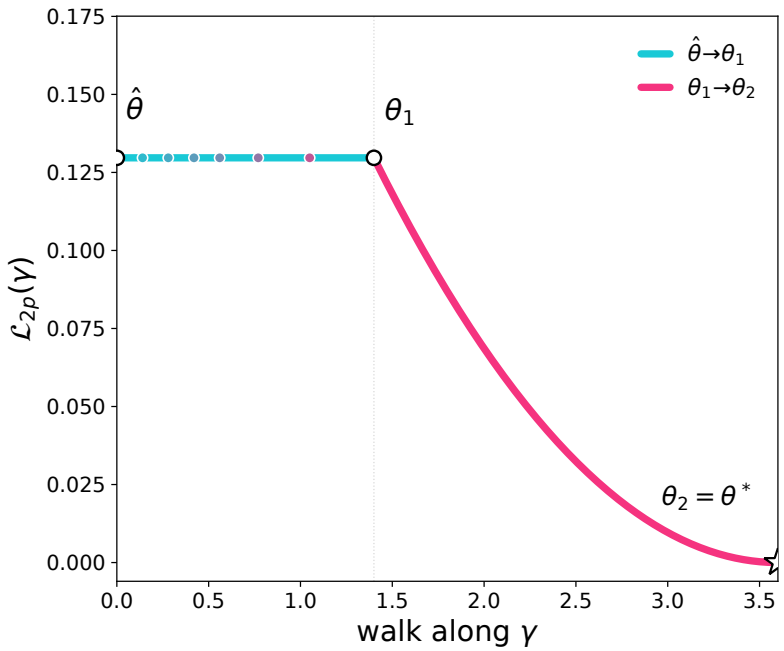


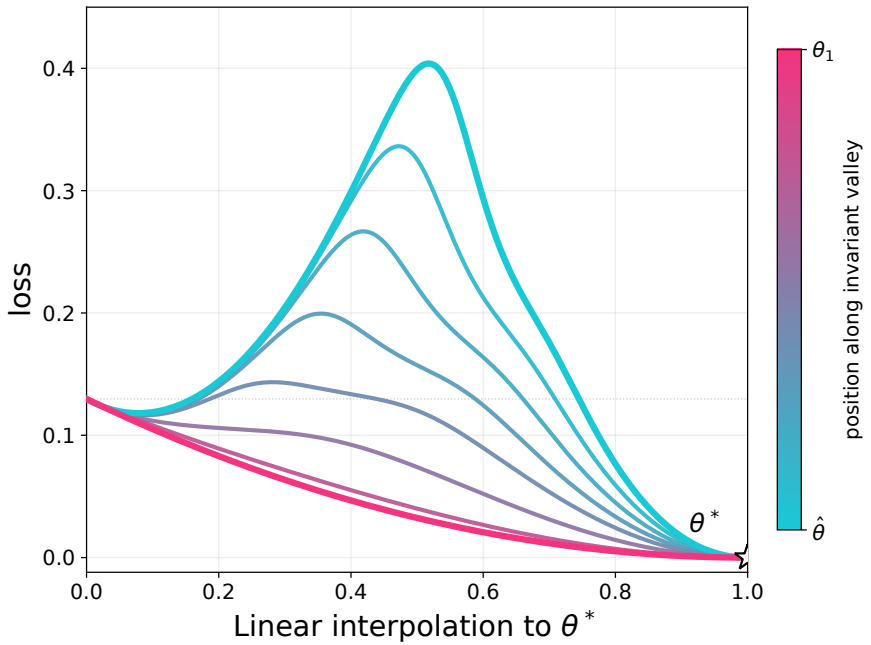


Sketch of Proof

Idea: Utilize degrees of freedom and weight sharing (convexity)

Walk segment	**Loss along path**	**Justification**
$\hat{\theta} \to \theta_1$	constant	same function
$\theta_1 \to \theta_2$	non-increasing	convexity + global min.





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Tight?

Question

Can we do better? E.g., $p + 1$?

Tight?

Question

Can we do better? E.g., $p + 1$?

Answer

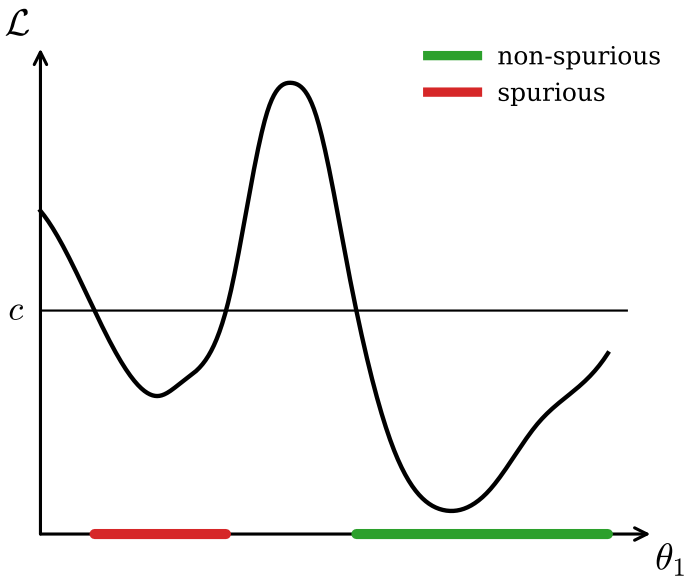
In general no.

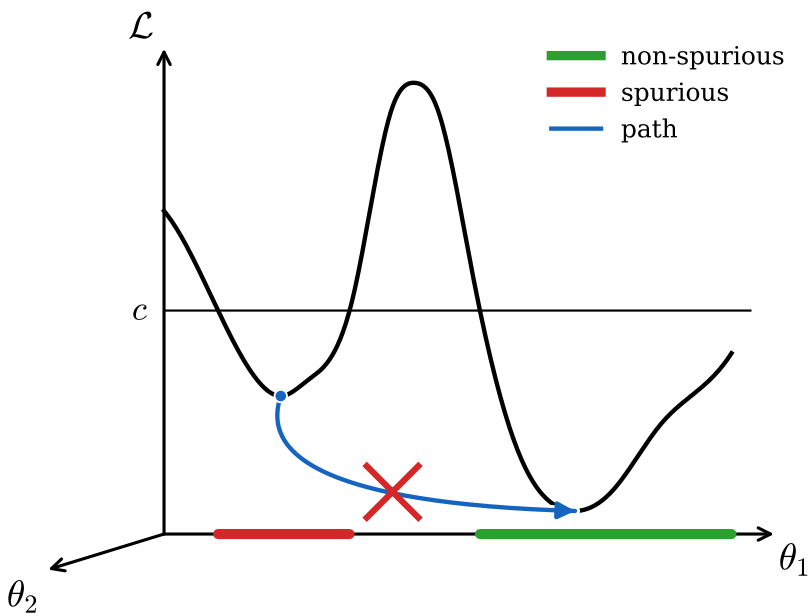
Tightness of Doubling

Theorem (informally)

There exists a functional loss such that:

- **Doubling** is **necessary** ($2p - 1$ not enough)
- Without doubling, **arbitrarily bad** valleys remain

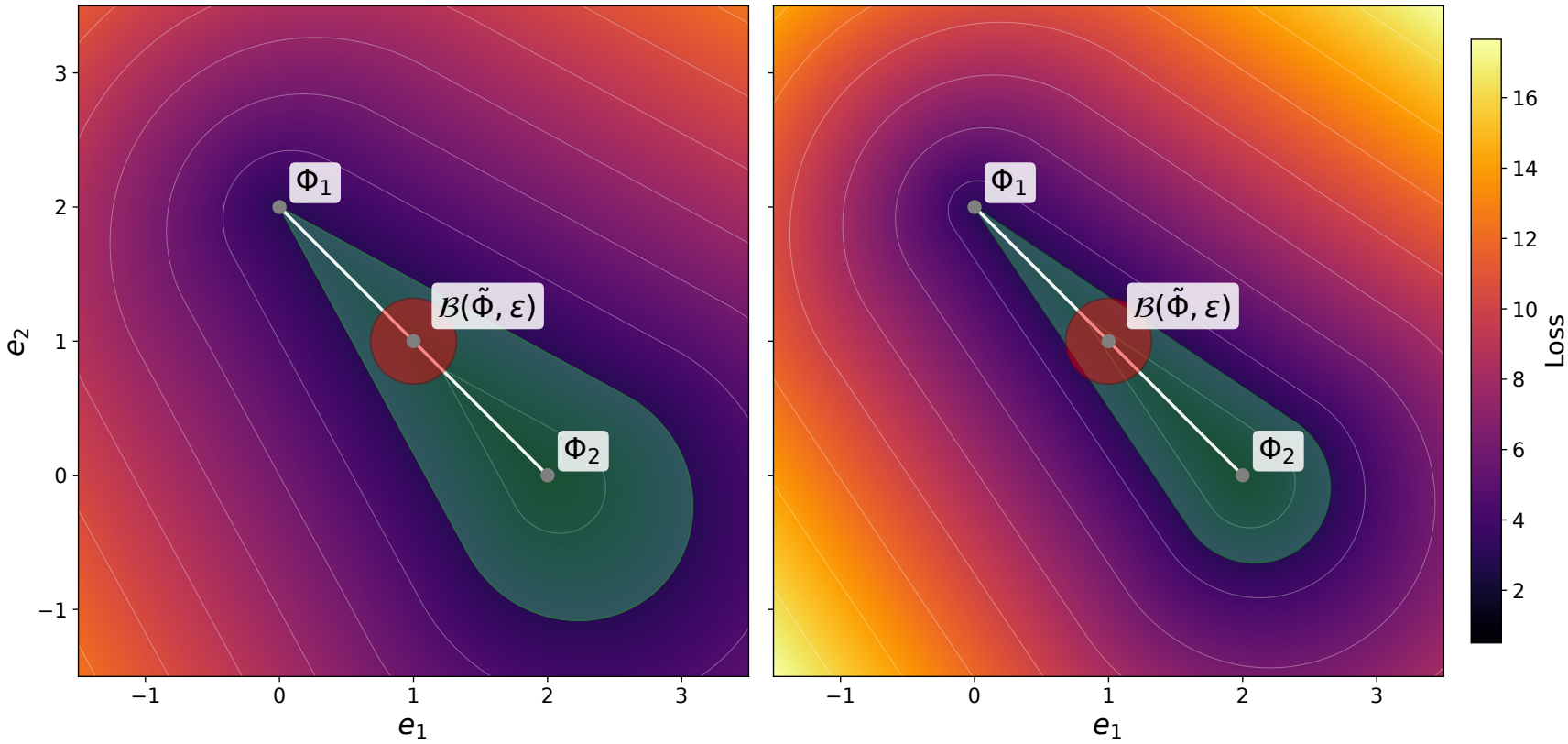




Sketch of Proof

Construct a functional loss such that:

- Red ball requires $\geq 2p$ neurons
- Unless we go through red ball $\rightarrow$ high loss



Non Adversarial Loss?

Question

Adversarial construction. What about more commonly used loss functions?

Non Adversarial Loss?

Question

Adversarial construction. What about more commonly used loss functions?

Answer

It is better

Expected Squared Distance

Expected squared distance:

$$l(\Phi) := \|\Phi - f^*\|_{L^2}^2, \qquad f^* \in \mathcal{F}$$

Theorem (informally)

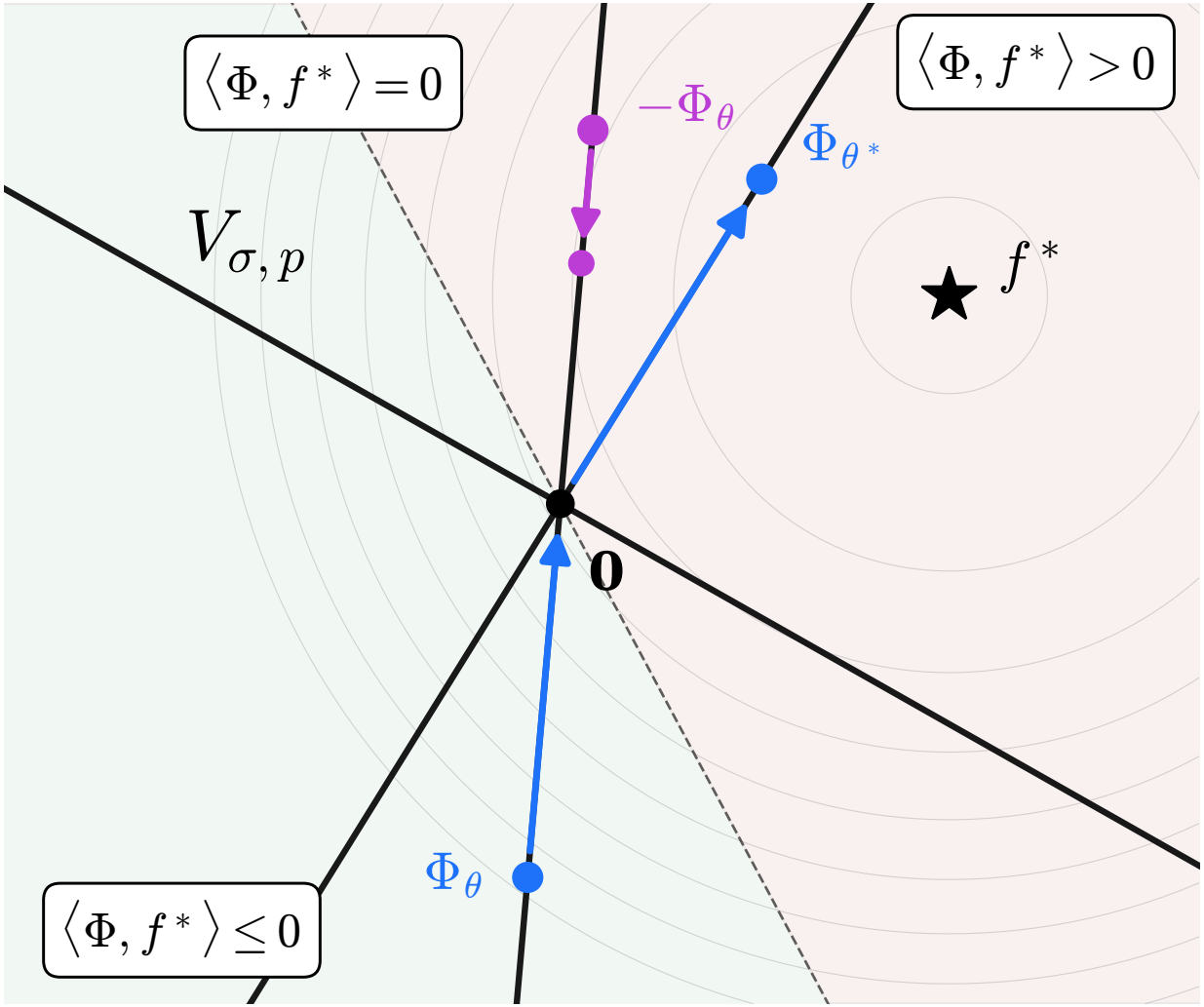
For the expected squared distance functional loss, and common initializations:

- Probability of escape $\geq 1/2$ even without overparam
- Descend path consists of three linear segments

$\Longrightarrow$ For commonly used methods, spurious valleys are not that bad

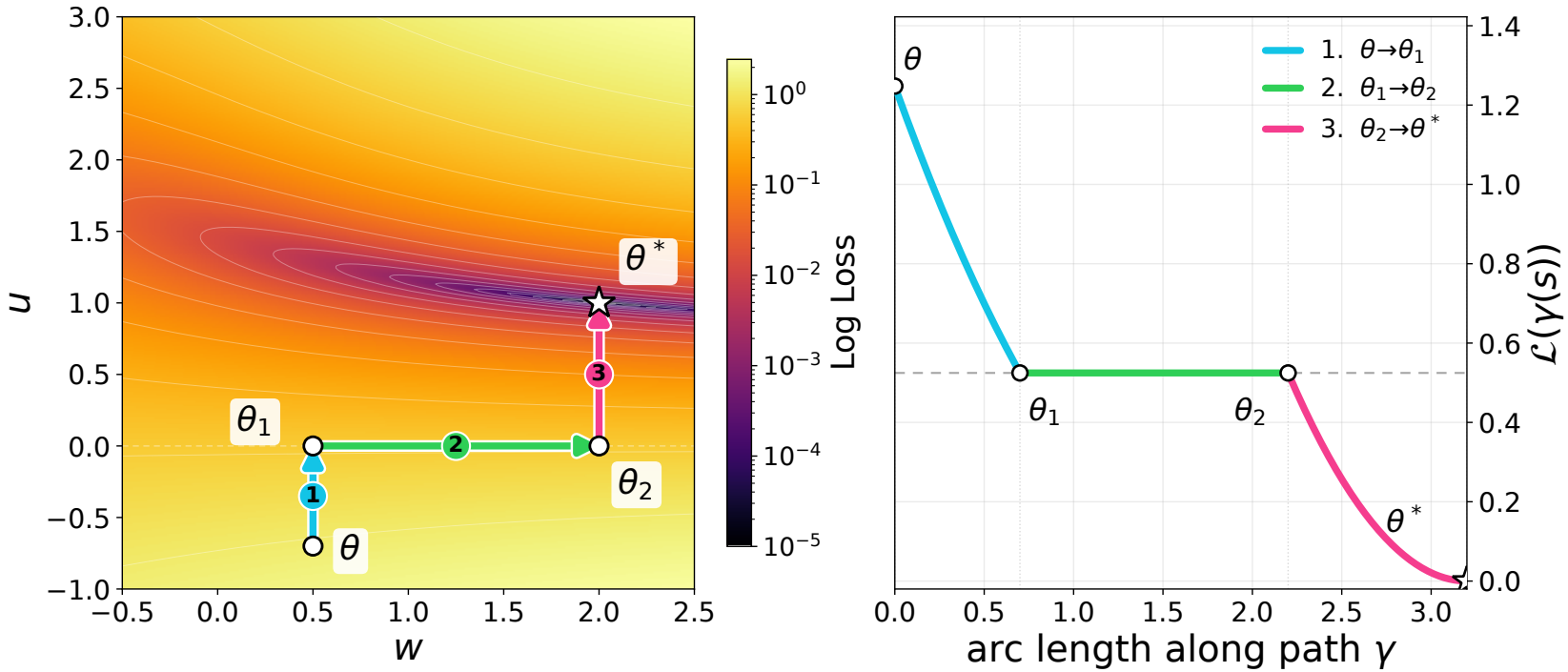
Sketch of Proof

In function space:



Sketch of Proof

In parameter space:



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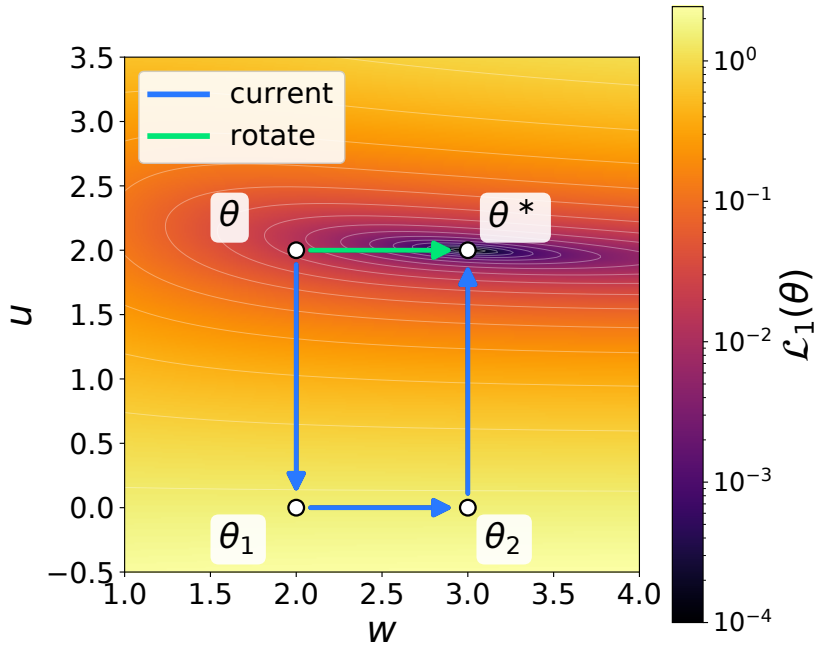
Recap

We have shown:

- Doubling is always enough to escape
- Doubling is necessary in general
- For expected squared distance: $\geq 1/2$ prob. of escape
  - Probability of escape monotonically increases via overparam.

No mention of:

- Optimization based methods
- Some different paths $\rightarrow$



Discussion

Generalization:

- General architecture: $\mathcal{N}(x; (U, W)) = U\varphi(x; W)$

Open Questions:

- Do optimization methods use these escape paths?
- Can optimization methods be developed to guarantee escape?
- Can we find different paths for faster escape?
- Ratio of newly appearing spurious valleys?

Thank you.

Assumptions for Main Result

- A1. $\sigma : \mathbb{R} \to \mathbb{R}$ is (non-polynomial) and continuous.
- A2. $l : \mathcal{F} \to \mathbb{R}$ is convex and continuous
- A3. For all $\theta \in \Theta_q$ the functional equivalence class $\mathcal{F}_{\mathcal{N}}(\theta, p)$ is path-connected when $p > q$.
- A4. In width p the global minimizer is attainable

$$\arg\min_{\Phi \in V_{\sigma,\infty}} (l(\Phi)) \cap V_{\sigma,p} \neq \emptyset$$

Remark

A3. and A4. can be relaxed.

Main Result

Theorem

Let Assumptions A1-A4 hold, $p \in \mathbb{N}_+$*, and* $\theta^* \in \arg\min_{\vartheta \in \Theta_p} \mathcal{L}_p(\vartheta)$*. For all* $\theta \in \Theta_p$*, and all* $\hat{\theta} \in \mathcal{F}_{\mathcal{N}}(\theta, 2p)$ *there exists a path* $\gamma : [0,1] \to \Theta_{2p}$ *that is non-increasing with respect to* $\mathcal{L}_{2p}$*, satisfying* $\gamma(0) = \hat{\theta}$ *and* $\gamma(1) \in \mathcal{F}_{\mathcal{N}}(\theta^*, 2p)$*.*

Assumptions for Tightness of Doubling

- A1. $\sigma : \mathbb{R} \to \mathbb{R}$ is non-polynomial and continuous.
- A5. For all widths $p \in \mathbb{N}_+$, there exists $\Phi \in V_{\sigma,p}$ and $\epsilon > 0$ with

$$\{f \in \mathcal{F} : \|\Phi - f\| \leq \epsilon\} =: \mathcal{B}(\Phi, \epsilon) \cap \overline{V_{\sigma,p-1}} = \emptyset$$

A5.: in width p there exists a network that cannot be arbitrarily well approximated with $p-1$ neurons.

[Venturi et al.(2019), Mondelli et al. (2018)]: similar results to A5.

Tightness

Theorem

Let A1 and A5 hold, let $p \in \mathbb{N}_+$, and let $M \in \mathbb{R}_+$.

Then there exist

- *convex, continuous functional loss l*
- *$\theta, \theta^* \in \Theta_p$, θ^* global minimizer*

such that the equivalence class $\mathcal{F}_{\mathcal{N}}(\theta, 2p-1)$ remains an arbitrarily bad spurious valley:

$$\mathcal{F}_{\mathcal{N}}(\theta, 2p-1) \subseteq \mathcal{V}_{2p-1}$$

and for any escape path γ, we have $\max_t \mathcal{L}_{2p-1}(\gamma(t)) \geq M$.

Expected Squared Distance

Theorem

Let Assumption A1 hold, and let $l(\Phi) = \|\Phi - f^*\|_{L^2}^2$ *for some* $f^* \in L^2([0,1]^n; \mathbb{R}^m)$*, and let* $\theta^* = (U^*, W^*) \in \arg\min_{\vartheta \in \Theta_p} \mathcal{L}_p(\vartheta)$*.*

Let μ *be a probability measure on* $(\Theta_p, B(\Theta_p))$ *under which* $U \overset{d}{=} -U$ *and* $U \perp\!\!\!\perp W$*, where* $\theta = (U, W) \sim \mu$*. Then*

$$\varrho_p(0) \geq \frac{1}{2}.$$

More precisely, every $\theta \in \Theta_p$ *satisfying* $\langle \mathcal{N}_p(\cdot; \theta), f^* \rangle_{L^2} \leq 0$ *is* 0*-escapable via a non-increasing path consisting of three explicit linear segments:*

$$(U, W) \;\to\; (0, W) \;\to\; (0, W^*) \;\to\; (U^*, W^*).$$